Novation Circuit Mono Station patch SysEx format

rev. 2020-06-30 reverse engineered by chaocrator

header (9 bytes)

offsets 00 to 08

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F0 00 20 29 01 61 00 00 00
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F0h	SysEx start
00h 20h 29h	Novation manufacturer ID
01h	Novation product type (synth)
61h	Novation product number (Circuit Mono Station)
00h	command (replace current patch; there are others – did not try them yet)
00h	location (Mono Station has the only location)
00h	reserved

patch name (16 bytes)

offsets 09 to 24

space symbol (20h) is used to pad unused symbol locations

patch data (100 bytes)

offsets 25 to 134

some Mono Station parameters have 0–255 value range (instead of 0—127).
so, they have sort of MSB/LSB.
what is referred as MSB is not real MSB, but can be 0 for values below
(on the left of) center or 40h / 64dec for center and above.

init hex & dec are the values of init patch.

offset	init hex	init dec	range	parameter
25	0	0		(probably category)
26	0	0		
27	0	0	0—1	env retrigger/legato
28	0	0		
29	0	0		
30	0	0		
31	0	0		
32	0	0		
33	0	0		
34	0	0		
35	0	0		
36	0	0		
37	0	0		
38	0	0		
39	0	0		
40	0	0		

offset	init hex	init dec	range	parameter
41	40	64	-6–+5	osc1 kbd octave
42	2	2	0–3	osc1 waveform
43	40	64	0–127	osc1 PW
44	40	64	0–127	osc1 coarse LSB
45	0	0	0/64	osc1 coarse MSB (0/40h)
46	32	50	0–100	osc1 fine LSB
47	0	0	0/64	osc1 fine MSB (0/40h)
48	1	1	0–3	osc1 range
49	0	0		
50	40	64	-6–+5	osc2 kbd octave
51	2	2	0–3	osc2 waveform
52	40	64	0–127	osc2 PW
53	40	64	0–127	osc2 coarse LSB
54	0	0	0/64	osc2 coarse MSB (0/40h)
55	32	50	0–100	osc2 fine LSB
56	0	0	0/64	osc2 fine MSB (0/40h)
57	1	1	0–3	osc2 range
58	0	0		
59	0	0	0–1	osc 1/2 sync
60	64	100		
61	7f	127	0–127	osc1 level LSB
62	40	64	0/64	osc1 level MSB (0/40h)
63	0	0	0–127	osc2 level LSB
64	0	0	0/64	//osc2 level MSB (0/40h)
65	0	0	0–127	sub osc level LSB
66	0	0	0/64	//sub osc level MSB (0/40h)
67	0	0	0–127	noise level LSB
68	0	0	0/64	noise level MSB (0/40h)
69	0	0	0–127	audio in level LSB
70	0	0	0/64	audio in level MSB (0/40h)
71	0	0	0–127	ring mod level LSB
72	0	0	0/64	ring mod level MSB (0/40h)
73	0	0	0–127	filter overdrive
74	0	0	0–127	filter reso
75	7f	127	0–127	filter cutoff LSB
76	40	64	0/64	filter cutoff MSB (0/40h)
77	0	0	0–2	filter shape (LP/BP/HP)
78	1	1	0–1	filter slope (12/24 dB)
79	0	0	0–3	filter bypass (0 – none, 1 – noise, 2 – osc2, 3 – noise & osc2)
80	7	7	0–7	filter key tracking
81	2	2	0–127	env atack
82	5a	90	0–127	env decay
83	7f	127	0–127	env sustain
84	28	40	0–127	env release
85	0	0	0–3	lfo waveform
86	22	34	0–127	lfo rate LSB
87	0	0	0/64	lfo rate MSB (0/40h)
88	0	0	0–34	lfo clock sync rate
89	0	0	0–1	lfo clock sync on/off
90	0	0	0–1	lfo key sync on/off
91	0	0	0–127	distortion amount
92	0	0	0–2	distortion type

offset	init hex	init dec	range	parameter
93	40	64	0–127	mod — env to osc1 pitch LSB
94	0	0	0/64	mod — env to osc1 pitch MSB (0/40h)
95	40	64	0–127	mod — env to osc1 PW
96	40	64	0–127	mod — env to osc2 pitch LSB
97	0	0	0/64	mod — env to osc2 pitch MSB (0/40h)
98	40	64	0–127	mod — env to osc2 PW
99	1	1		mod — env to amp (**boolean — 0 or 1**)
100	40	64	0–127	mod — env to cutoff LSB
101	0	0	0/64	mod — env to cutoff MSB (0/40h)//
102	40	64	0–127	mod — env to distortion
103	40	64	0–127	mod — env to aux CV
104	40	64	0–127	mod — lfo to osc1 pitch LSB
105	0	0	0/64	mod — lfo to osc1 pitch MSB (0/40h)//
106	40	64	0–127	mod — lfo to osc1 PW
107	40	64	0–127	mod — lfo to osc2 pitch LSB
108	0	0	0/64	mod — lfo to osc2 pitch MSB (0/40h)//
109	40	64	0–127	mod — lfo to osc2 PW
110	40	64	0–127	mod — lfo to amp
111	40	64	0–127	mod — lfo to cutoff LSB
112	0	0	0/64	mod — lfo to cutoff MSB (0/40h)//
113	40	64	0–127	mod — lfo to distortion
114	40	64	0–127	mod — lfo to aux CV
115	40	64	0–127	mod — seq to osc1 pitch LSB
116	0	0	0/64	mod — seq to osc1 pitch MSB (0/40h)//
117	40	64	0–127	mod — seq to osc1 PW
118	40	64	0–127	mod — seq to osc2 pitch LSB
119	0	0	0/64	mod — seq to osc2 pitch MSB (0/40h)//
120	40	64	0–127	mod — seq to osc2 PW
121	40	64	0–127	mod — seq to amp
122	40	64	0–127	mod — seq to cutoff LSB
123	0	0	0/64	mod — seq to cutoff MSB (0/40h)//
124	40	64	0–127	mod — seq to distortion
125	40	64	0–127	mod — seq to aux CV
126	40	64	0–127	mod — vel to osc1 pitch
127	40	64	0–127	mod — vel to osc1 PW
128	40	64	0–127	mod — vel to osc2 pitch
129	40	64	0–127	mod — vel to osc2 PW
130	40	64	0–127	mod — vel to amp
131	40	64	0–127	mod — vel to cutoff
132	40	64	0–127	mod — vel to distortion
133	40	64	0–127	mod — vel to aux CV
134	0d	13	0–31	selected routing (see table in MIDI manual)

end of SysEx byte (F

offset 135